

Work Instruction — Part 45621-B OP30 — Finish 5-Axis Mill on DMU-14

Rev C · effective 2025-03-10. Operation OP30 (finish mill all pocket features) on Inconel 718. Backup machine `DMU-22` may be used; parameters identical.

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--- METADATA ---
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owner: M. Tanaka, Process Engineering — Wichita
part_numbers: [45621-B]
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machine_ids: [DMU-14, DMU-22]
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fault_codes: []
materials: [MAT-INC718]
standards: []
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1. Setup

- 1. Verify machine is at operating temperature (warm-up cycle complete — see `PDM-MAN-DMU14-007` §4).
- 2. Load fixture `FIX-45621B-OP30-01`. Verify pin location with go/no-go pin.
- 3. Load OP20 part (datum A flat & verified). Confirm seating with feeler gauge (0.05 mm shall not pass).
- 4. Load tools per tooling sheet `PLM-T00L-45621B-OP30` . Verify tool table active.
- 5. Run program `045621B_OP30_REVC` .

2. Cutting parameters

Sub-op	Tool	Spindle (rpm)	Feed (mm/min)	Depth of cut (mm)	Coolant
2-1: Pocket roughing	T05 — Ø 10 mm 4-flute carbide (AlTiN)	1,800	360	axial 1.5, radial 4.0	20 bar through-spindle

2-2: Pocket semi-finish	T06 — Ø 8 mm 4-flute carbide	2,400	320	axial 0.5, radial 0.4	20 bar
2-3: Pocket finish	T07 — Ø 6 mm 6-flute ball-nose carbide	3,200	250	axial 0.15, radial 0.2	20 bar

These parameters are **validated within the OEM machine envelope** documented in PDM-MAN-DMU14-007 §2 (12,000 rpm and 15,000 mm/min cutting feed are the OEM limits — well above the values used here). Tool-wear limit and material-specific guidance: PLM-PS-INC718-0042 .

3. In-process checks

1. First piece: full CMM check at OP40.
2. Every 10 pieces: visual check of T07 ball-nose insert. Replace if flank wear exceeds $VB_{max} = 0.20$ mm or if surface finish trends above $Ra\ 0.4\ \mu m$.
3. Every tool change: log time in DMU14_TOOL_LIFE spreadsheet. Cumulative T07 cut time shall not exceed 60 min between changes (per PLM-PS-INC718-0042).

4. Safety and PPE

- Cutting fluid is EcoCut 7700 (see SDS ERP-SDS-ECOCUT7700). PPE: safety glasses, nitrile gloves when topping up.
- Door interlock must be active during cutting. Do not override.
- For any tool-change or fixture-cleaning task requiring a hand inside the work envelope, follow the cell LOTO procedure.

Document control / sign-off

Role	Name	Signature	Date
Document Owner	M. Tanaka, Process Engineering — Wichita	(on file)	2025-03-10
Approver	M. Tanaka, Process Engineering — Wichita	(on file)	2025-03-10